Project Design Phase
Solution Architecture

Date	26 February 2026
Team ID	699d5fc83074df695bbbb0c1
Project Name	AI Financial Advisor
Maximum Marks	4 Marks

Solution Architecture:

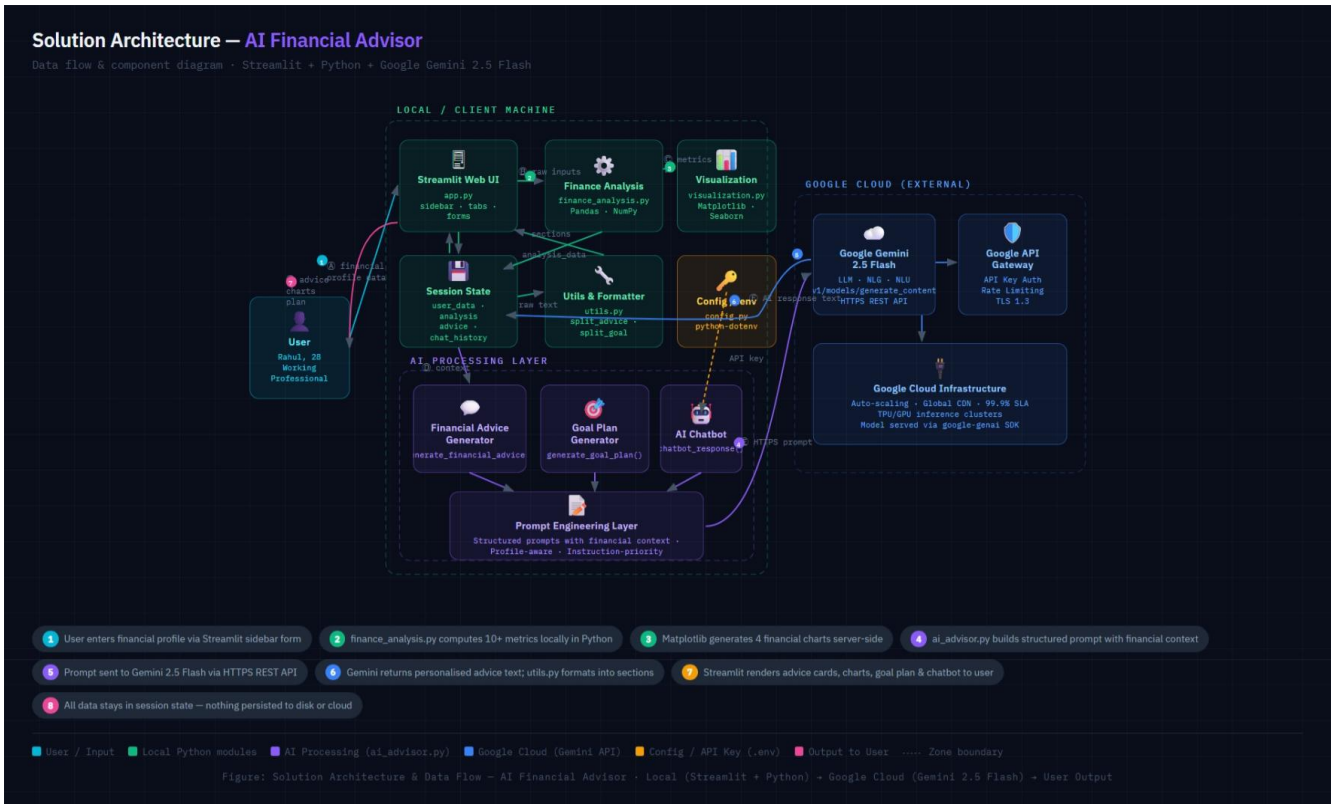

Figure 1: Architecture and data flow of the AI Financial Advisor web

3 Infrastructure Zones (dashed boundaries)

Zone	Color	Components inside
Local / Client Machine	Green dashed	Streamlit UI, Session State, finance_analysis.py, utils.py, visualization.py, config/.env
AI Processing Layer (inside Local)	Purple dashed	Financial Advice Generator, Goal Plan Generator, AI Chatbot, Prompt Engineering Layer
Google Cloud (External)	Blue dashed	Gemini 2.5 Flash, API Gateway (Auth + TLS), Google Cloud Infrastructure

14 Component Nodes — each color-coded by role (user, local, AI, cloud, config, output)

Numbered Arrow Flow (1→8)

1. User inputs financial profile via sidebar
2. finance_analysis.py computes 10+ metrics locally
3. Matplotlib generates 4 charts server-side
4. ai_advisor.py builds structured prompt
5. Prompt sent to Gemini via HTTPS REST
6. Gemini returns text → utils.py formats sections
7. Streamlit renders advice, charts, goal plan to user
8. All data stays in session state — nothing persisted

The dotted arrow from config.py shows the API key injection path, and the return path from Gemini back to session state is clearly traced — matching the expected format for IBM/SmartBridge solution architecture documentation.